

Abstraction Calculus (AC) — Minimal Specification v0.2

Draft for review

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Abstract

This document specifies a minimal, domain-agnostic calculus for building towers of abstraction. A *primitive* is not an atom but a stance: a universe $\mathcal{U}$, a valuation codomain $\mathcal{B}$ (with a declared “mode”), and a context-indexed *lens* $L_C : \mathcal{U}_C \rightarrow \mathcal{B}_C$ that extracts the basic truths of interest. Abstraction is the kernel quotient of L_C . Re-priming treats the first abstraction’s carrier as the next universe, possibly changing codomain via a functor and a new lens. We include rules for contexts, optional set-like operations (when $\mathcal{B}$ supports them), safety constraints for reflection, and minimal policies for stopping/loop-avoidance. This is intended to be sufficiently precise to be implemented or formalized further while remaining compact.

1 Preliminaries: Contexts and Universes

- A set of contexts $(\text{Ctx}, \preceq)$ forms a preorder. Intuitively, $C' \preceq C$ means C' is a restriction/coarsening of C .
- Each $C \in \text{Ctx}$ determines (possibly different) carriers $\mathcal{U}_C$ and codomains $\mathcal{B}_C$.
- For $C' \preceq C$ there are *restriction maps* $r_{C \rightarrow C'}^U : \mathcal{U}_C \rightarrow \mathcal{U}_{C'}$ and $r_{C \rightarrow C'}^B : \mathcal{B}_C \rightarrow \mathcal{B}_{C'}$.
- **Naturality.** A lens (Def. 1) must commute with restriction: $r_{C \rightarrow C'}^B \circ L_C = L_{C'} \circ r_{C \rightarrow C'}^U$.
- **Reconciliation (meet).** When combining objects from C_1 and C_2 , we require a greatest lower bound $C_{12} = C_1 \wedge C_2$ where defined. All operations must be interpreted in C_{12} . If the meet does not exist, the operation is undefined.

- **Universe levels.** To avoid size issues, we work in a fixed Grothendieck universe level; carriers may be large but remain in that universe.

2 Primitives and Modes

Definition 1 (Primitive). *A primitive at context C is a quadruple $\Pi = (\mathcal{U}, \mathcal{B}, L_C, \text{mode})$ where:*

- (i) $\mathcal{U}_C$ is the universe of objects of interest in context C ;
- (ii) $\mathcal{B}_C$ is the valuation codomain in C , equipped with whatever algebraic structure is declared (e.g. order, lattice operations);
- (iii) $L_C : \mathcal{U}_C \rightarrow \mathcal{B}_C$ is a lens (observation map) satisfying naturality in C ;
- (iv) $\text{mode}(\mathcal{B}) \in \{\text{descriptive, normative, operational}\}$ labels the semantics of values. Certain compositions are forbidden unless an explicit mode-bridge is provided.

Remark 1 (Deterministic, partial, and probabilistic lenses). *We allow deterministic lenses $L_C : \mathcal{U}_C \rightarrow \mathcal{B}_C$, partial lenses $L_C : \mathcal{U}_C \rightarrow \mathcal{B}_C$ (with a designated bottom $\perp \in \mathcal{B}_C$), and probabilistic lenses $L_C : \mathcal{U}_C \rightarrow \Delta(\mathcal{B}_C)$ into distributions. All definitions below lift pointwise; for partiality we work in a suitable Kleisli category (error/option monad).*

3 Abstraction as Kernel Quotient

Definition 2 (Observational equivalence). *For $\Pi = (\mathcal{U}, \mathcal{B}, L_C, \text{mode})$, define $x \sim_C y$ iff $L_C(x) = L_C(y)$. Let $q : \mathcal{U}_C \rightarrow A_1$ be the quotient map to the set of equivalence classes $A_1 := \mathcal{U}_C / \sim_C$.*

Definition 3 (Abstraction operator α). *The abstraction of Π is*

$$\alpha(\Pi) := (A_1, \mathcal{B}, \hat{L}_C, \text{mode}), \quad \hat{L}_C([x]) := L_C(x).$$

Lemma 1 (Well-definedness). *$\hat{L}_C$ is well-defined: if $[x] = [y]$ then $\hat{L}_C([x]) = \hat{L}_C([y])$.*

Proposition 1 (Idempotence). *α is idempotent: $\alpha \circ \alpha = \alpha$ (up to canonical isomorphism of carriers).*

Remark 2 (Non-triviality guard). *To avoid collapse, define a dispersion/complexity score $H(L_C)$ (e.g. entropy of the class distribution or MDL bits saved). A system may forbid applying α when H would drop below a floor unless a downstream score (Sec. 9) is expected to increase.*

4 Re-Priming and Change of Codomain

There are two orthogonal ways to proceed after α :

- **New lens on the new carrier.** Choose B'_C and a lens $L'_C : A_1 \rightarrow B'_C$. Define the *re-primed primitive*

$$\text{Reprime}(\Pi, L'_C) := (A_1, B', L'_C, \text{mode}(B')).$$

- **Change of base along a functor.** If $G : \mathcal{B} \rightarrow \mathcal{B}'$ is a (contextual) functor, one may change codomain to $G(\mathcal{B})$ while post-composing the lens: $G \circ L_C : \mathcal{U}_C \rightarrow G(\mathcal{B}_C)$.

Proposition 2 (Commutation (factorization / Beck–Chevalley)). *Let $q : \mathcal{U}_C \rightarrow A_1$ be the quotient for L_C . If the composite $L'_C \circ q$ factors through L_C (i.e. is constant on $\sim_C$ -classes), then abstraction followed by re-priming equals re-priming followed by abstraction (up to isomorphism). More generally, when re-priming is expressed as change of base along G , commutation holds under a Beck–Chevalley condition for the square induced by q and G .*

5 Morphisms and the Category Abs

Definition 4 (Morphisms). *A morphism $\Phi : \Pi \rightarrow \Pi'$ between primitives $\Pi = (\mathcal{U}, \mathcal{B}, L_C, \text{mode})$ and $\Pi' = (\mathcal{U}', \mathcal{B}', L'_{C'}, \text{mode}')$ consists of $(f : \mathcal{U}_C \rightarrow \mathcal{U}'_{C'}, g : \mathcal{B}_C \rightarrow \mathcal{B}'_{C'})$ and a context map $\rho : C \rightarrow C'$ such that $g \circ L_C = L'_{C'} \circ f$ and naturality holds with respect to restrictions along ρ . Composition is pointwise; identities are $(\text{id}, \text{id}, \text{id})$.*

Proposition 3. *With the above morphisms, primitives form a category Abs. The operator α extends to a functor $\alpha : \text{Abs} \rightarrow \text{Abs}$ that is idempotent (a reflector).*

6 Functionization (Systems of Functions)

Given $\Pi = (\mathcal{U}, \mathcal{B}, L_C, \text{mode})$, let F be the algebra freely generated by a finite signature of basic features/predicates derived from L_C under composition and products. Equip F with a lens $L_C^F : F \rightarrow \mathcal{B}_C$ (e.g. utility, stability). Then $(F, \mathcal{B}, L_C^F, \text{mode})$ is a primitive and the calculus iterates. Finite presentation avoids uncontrolled blowup.

7 Optional Set-like Instance

Assume each $\mathcal{B}_C$ is a De Morgan algebra $(\wedge, \vee, \neg)$ and each membership lens $M_S : \mathcal{U}_C \rightarrow \mathcal{B}_C$ is a homomorphism.

$$\begin{aligned}(M_{A \cap B})(x) &:= M_A(x) \wedge M_B(x), \\ (M_{A \cup B})(x) &:= M_A(x) \vee M_B(x), \\ (M_{A^c})(x) &:= \neg M_A(x).\end{aligned}$$

Theorem 1 (De Morgan). $(A \cap B)^c = A^c \cup B^c$ and $(A \cup B)^c = A^c \cap B^c$ (equality pointwise in $\mathcal{B}_C$).

Remark 3 (Context-relative complement). *Complements are taken relative to $\mathcal{U}_C$; cross-context complements are disallowed unless both terms are transported to a common meet context $C_1 \wedge C_2$.*

8 Safety: Reflection and Partiality

- **Stratified reflection.** Lenses at level k (on carrier A_k) may *evaluate* only objects at strictly lower levels. This prevents liar/Russell-type paradoxes while allowing meta-reasoning by level-shifts.
- **Guarded recursion.** Same-level self-reference is permitted only through contractive/guarded operators in an ordered or metric setting; admissible fixed points are then unique.
- **Partial evaluation.** Downward “evaluation” maps are modeled as Kleisli morphisms for an error/option monad; failures propagate explicitly.

9 Policies: Stopping and Loop-Avoidance

Attach to each level a score J_k in common units (e.g. per-datum MDL bits saved, predictive stability, or cross-observer agreement), and monitor a dispersion score $H(L_C)$ for non-triviality.

- **Loop-break.** Stop lifting when ΔJ_k falls below a threshold or contradictions exceed a tolerance.
- **Loop-avoid.** Learn policies over operator sequences $(\alpha, \text{Reprime}, \text{compose})$ to avoid historically futile paths.

10 Diagnostics Checklist (per instantiation)

1. What is $\mathcal{B}$ and its mode? What algebraic structure does it carry?
2. Is L_C deterministic, partial, or probabilistic? How is uncertainty handled?
3. Does a meet context exist for the operations you need? Are restrictions natural?
4. Will α over-collapse? Check $H(L_C)$ and expected ΔJ_k .
5. Are set-like operations legitimate (homomorphism into a De Morgan/Heyting algebra)?
6. Do re-priming steps commute with α (factorization / Beck–Chevalley)?
7. If combining abstraction chains, do you have a pullback/common refinement of carriers?
8. Are comparison metrics (if any) monotone under abstraction?

11 Micro-Examples

Example 1 (Valence/EOR). U_C : *experiences*; $\mathcal{B}_C$: *valence scale*; L_C : *felt valence*. α *yields patterned-feeling classes*; *re-priming to norms (ethics) equips A_1 with a new lens*. *Safety rules prevent self-referential collapse*.

Example 2 (Counting/Mathematics). U_C : *finite collections*; $\mathcal{B}_C = \mathbb{N}$; L_C : *“how many?”*. α *collapses to equinumerosity classes*; *re-priming with structure-sensitivity lenses moves to algebraic carriers (monoids/groups)*.

Example 3 (Fashion/Taste). U_C : *outfit* $\times$ *context* $\times$ *observer*; $\mathcal{B}_C = \{\text{like}, \text{dislike}\}$; L_C : *instant taste*. α *gives style classes*; *re-priming yields “trend impact”*; *set-like ops are available if $\mathcal{B}_C$ is given an appropriate algebra*.

A Selected Proof Sketches

Proof of Lemma 1. If $[x] = [y]$ then $x \sim_C y$ so $L_C(x) = L_C(y)$ by definition; hence $\hat{L}_C([x])$ is independent of representatives. $\square$

Idempotence of α (Prop. 1). The kernel of $\hat{L}_C$ is trivial on A_1 because $\hat{L}_C([x]) = L_C(x)$; the induced quotient $A_1/\sim$ is canonically isomorphic to A_1 . $\square$

Commutation (Prop. 2). If $L'_C \circ q$ factors through L_C , then there exists $h : \text{im}(L_C) \rightarrow B'_C$ with $L'_C \circ q = h \circ L_C$. The first abstraction identifies precisely $\ker L_C$ -classes; the two routes then define the same map on classes. The functorial statement follows from a Beck–Chevalley condition on the square formed by q and G . $\square$

De Morgan (Theorem in Sec. 7). Pointwise in $\mathcal{B}_C$, using De Morgan algebra axioms: $\neg(a \wedge b) = \neg a \vee \neg b$ and $\neg(a \vee b) = \neg a \wedge \neg b$. $\square$